Enhancing Psychosocial Counselling with AI: A Multifaceted Support System for Professionals

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Abstract. The expanding field of psychosocial online email counselling has created a demand for advanced tools to support counsellors in managing diverse client interactions. This paper introduces CAIA (Counsellor Artificial Intelligence Assistant), a comprehensive AI-driven system currently under active development, designed to condense and present information to counsellors in a structured manner. CAIA integrates several AI-based features, including Automatic Subject Generation, Case Summarisation, Timeline Management, Social Network Graph creation, and Case Classification. The features are designed to assist the counsellors, allowing them to engage more deeply with each case while ensuring that the client's needs remain paramount. Each feature presented in the paper addresses a specific problem in psychosocial online counselling, proposes a solution, discusses a technical implementation approach, and highlights the challenges encountered. Additionally, the paper discusses ethical considerations associated with an AI integration.

Keywords: AI · Large Language Models · Online Counselling · Psychosocial Support · Case Management · Information Processing

1 Introduction

The provision of psychosocial online email counselling, which offers an easily accessible, free, and anonymous alternative to face-to-face support for social and psychological problems [22], is undergoing a period of rapid expansion [30]. During periods of crisis, there is an increase in demand for psychosocial support [35, 36]. For example, the demand for online counselling services increased significantly during the pandemic [20, 24]. These circumstances present considerable challenges for counsellors.

A noteworthy challenge is the inconsistent training among counsellors. The current academic training often uses role-playing exercises, which may foster a reliance on the respective role-playing partner [31]. Furthermore, it can be challenging to understand a single client's concerns in the absence of non-verbal communication cues such as gestures and facial expressions [8, 21]. This makes it challenging to fully comprehend the client's concerns and demonstrate empathy [17]. Additionally, counsellors must manage numerous concurrent conversations, making it difficult to maintain an accurate record of individual cases [37]. Furthermore, the field is experiencing a shortage of qualified personnel [38]. The

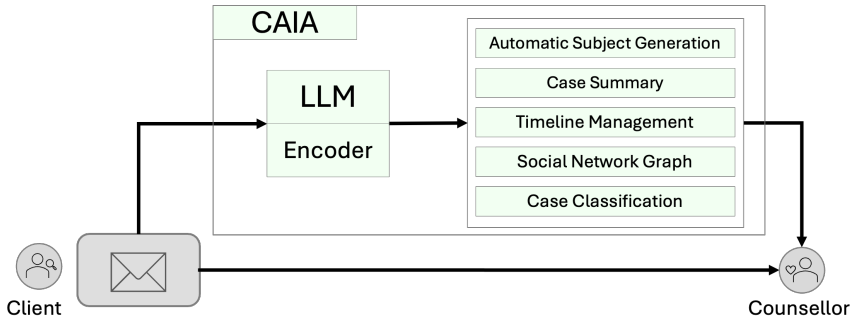

Fig. 1. CAIA Information Processing: The system leverages several AI supported features to provide counsellors with a complete and structured overview of client interactions alongside the original emails.

complexity and urgency of these problems require innovative solutions that may lie in the use of artificial intelligence (AI) [26] in general and large language models (LLMs) in specific [11]. The utilisation of AI and LLMs can serve to mitigate these issues by facilitating the visualisation and condensation of the available information, given that they can only work with the information provided by clients in the form of emails. AI represents a promising approach to improving mental health care, as evidenced by the growing popularity of this technology [12, 14, 9]. However, due to the sensitive nature of counselling data, it is imperative that these AI systems operate locally within the secure environment of the counselling organisations. Using third-party AI services is not viable due to privacy concerns and the potential for data breaches. This paper introduces the conceptual framework of CAIA (Counsellor Artificial Intelligence Assistant), a system in active development, designed to support professional counsellors in the extraction and processing of information (Fig. 1). CAIA incorporates a number of innovative AI-driven features designed to address the challenges inherent to psychosocial online counselling. The Automatic Subject Generation facilitates the rapid identification and organisation of cases. The Case Summarisation tool provides concise and relevant summaries of past interactions. The Timeline Management enables the accurate tracking of significant events and interactions over time. The Social Network Graph offers a visual representation of the client's social environment. Finally, Case Classification assists in the efficient categorisation of cases. Each feature is recalculated with every email exchanged to ensure up-to-date information.

After a brief overview of related work, we provide a detailed explanation of each feature, addressing the challenges and potential solutions. A particular focus of our work is on ethical considerations, which are discussed in Section Four. We conclude the paper with an overview of the current state of the implementation and an outlook on our future work.

2 Related Work

AI can be employed in a multitude of ways in the field of online counselling. This section will discuss three such applications.

2.1 AI as counsellor

The history of chatbots is closely linked to mental health. The first chatbot in this field was ELIZA, a chatbot developed in 1966 by Joseph Weizenbaum [39]. ELIZA, which was intended to simulate a psychotherapist according to Carl Rogers, used a rule-based approach to conduct simple but effective dialogues, laying the foundation for many subsequent chatbots. Another important step was the introduction of A.L.I.C.E. (Artificial Linguistic Internet Computer Entity). A.L.I.C.E. was introduced by Richard Wallace in 1995 and is a technically advanced version of ELIZA [32]. Since then, numerous publications have appeared in the field of chatbots for counselling, particularly in the area of psychotherapy and mental health [40]. Since ELIZA and A.L.I.C.E., research has made great progress in the development of chatbots that can be used in psychosocial counselling. Newer systems such as Chatcounsellor [25], Serena [5] and Psy-LLM [24] represent further developments of this technology. These systems aim to conduct realistic counselling sessions and simulate specific therapeutic techniques. These systems are not yet able to autonomously replace the role of humans as counsellors or therapists [11, 23]. However, research has shown that chatbots can improve access to counselling as they are available around the clock [13]. These systems can serve as a first point of contact to support people with mental health problems and guide them on the path to conventional therapy.

2.2 AI as Client for Educational Role-Play

Research demonstrates that AI can effectively simulate clients in training scenarios, providing counsellors with a risk-free environment to develop their skills. Maurya and DeDiego [28] discuss the integration of AI chatbots like ChatGPT in counsellor education, highlighting their ability to simulate diverse psychological conditions and behaviors. This method provides a controlled and varied learning environment, overcoming logistical challenges of using human actors. Maurya [27] further explores ChatGPT's effectiveness in role-play, noting its realistic emotional responses and cultural sensitivity. While AI chatbots offer valuable training benefits, they occasionally produce overly idealized responses due to their reliance on training data patterns. Rudolph et al. [31] introduced 'VirCo' (Virtual Client for Online Counselling), an AI system using large language models to generate realistic client responses. This tool enhances traditional role-playing by offering independent practice opportunities and improving training authenticity through detailed persona descriptions and session transcripts.

2.3 AI as Counsellor Support

Several studies suggest that AI or LLMs can effectively support counsellors in psychosocial online counselling [1, 4, 12, 33]. The role of AI in supporting advisors in daily practice is a relatively new field of research [16]. AI systems can relieve human advisors by taking over administrative tasks, summarising information, providing decision support and offering assistance with the advisor's response. In a study [16], a system called Reply+ was developed, which revises the advisor's message on request. In another study by [19], a system called CARE was developed which, like Reply+, improves the response from unprofessional advisors. HAILEY from [34] is also a system that helps non-professional counsellors to write messages to people seeking advice. The system supports the counsellor in facilitating the expression of empathy in supportive conversations in text form. In contrast to other systems that focus on improving the counsellor's response, this paper highlights and proposes several ways in which AI can improve the counselling process and support the counsellor.

3 Conceptual Framework of CAIA

In this section, we present CAIA, a system designed to systematically support professional counsellors in their work with clients in online counselling. The following subsections examine multiple AI-based features, namely Automatic Subject Generation, Case Summarisation, Timeline Management, Social Network Graph creation, and Case Classification. These features leverage AI technologies, such as LLMs and transformer-based classifiers, to visualize information and present it in a condensed manner, assisting counsellors in managing and understanding client interactions. Additionally, CAIA offers the technical capability to make changes to features and store these modifications for future AI training.

When employing LLMs to address specific tasks, a structured prompt approach can be designed as follows:

LLM Prompt Template for Task Structuring

- **LLM Role Definition** *Defines the role of the LLM in a specific situation.*
- **Context** *Provides contextual information and specific requirements to ensure appropriate responses.*
- **Task Definition** *Describes what the language model is supposed to do.*
- **Input Format** *Explains the structure of the input data.*
- **Output Format** *Explains the expected output format.*
- **Avoid Formalities** *Specifies wording to be avoided.*
- **Data** *The data to be processed should be inserted into this section in the formatting specified above.*
- **Role Reminder** *Reminds the model of its role during the task.*
- **Task Reminder** *Reminds the model of the specific task to maintain focus and goal orientation.*

3.1 Automatic Subject Generation

The work of online counsellors is characterized by the need to manage a large number of cases simultaneously. It is important to quickly navigate through different cases and identify a specific case by its heading or precise subject line.

The considerable number of cases that must be processed concurrently presents a challenge for counsellors, who may find it difficult to navigate the system and immediately identify a client's specific case. The initial subject lines of clients are often brief and generalised, employing terms such as "grief" or "help". These subject lines provide minimal helpful information for quick orientation.

As a potential solution, CAIA utilises an LLM to analyse the initial message and generate an additional, contextual subject line that more accurately captures the essence of the client's message than the subject line originally chosen by the client. The dual subject line system is not intended to supplant the client's voice, but rather to enhance the counsellor's approach.

The challenge is to instruct the LLM to produce a concise subject line which is free from any supplementary text and unnecessary elements such as quotation marks or prefixed words like "Subject". Thus, the prompt must be carefully designed and a post-processing algorithm should be implemented to reinforce that only a single subject line without any colons or quotation marks is generated.

3.2 Case Summarisation

Online mental health counsellors frequently encounter the challenge of managing a substantial number of cases [2]. It can be challenging to recall specific case details if there has been a prolonged period of inactivity or if the case has been transferred from another legal professional who is currently on leave.

Traditional methods of reviewing case histories involve reading through lengthy transcripts of previous sessions or even the whole email conversation. The process is time-consuming and prone to oversights, where important details can be overlooked, especially under time pressure.

The summarisation capabilities of LLMs have already been successfully validated in a variety of general domains [10]. LLMs can create concise yet comprehensive summaries of complex case interactions through proper prompting. The summaries focus on the essential points and topics so that advisors can quickly familiarise themselves with the key aspects of each case without having to search through extensive notes or recordings. The development of this AI-supported function can be achieved using an LLM that has been trained in the language in which the emails are written, as its summarisation capabilities are sufficient to implement this feature. Fine-tuning on existing counselling data can be applied to further improve performance.

One of the principal challenges is to guarantee that the LLM produces summaries that are both accurate and easy to comprehend by the counsellor. This requires well-crafted prompts to guide the LLM on what information to extract and how to present it concisely. Furthermore, the context window of the LLM must be sufficiently expansive to encompass the entirety of the communication.

Email conversations in counselling often span several months with intensive exchanges, which can exceed the context window limits. Open-source models typically have context windows ranging from 4,096 to 32,768 tokens. Since LLMs must be locally hosted to ensure data privacy, resource efficiency is crucial. Thus, models with larger context windows like GPT-4 and Claude are not an option in this setting.

While techniques such as rope scaling can adjust the context window, an alternative approach is employed. When the email thread becomes too lengthy, the oldest emails are removed until the most recent emails and the current summary fit within the context window, ensuring that the most recent information is always included.

3.3 Timeline Management

In asynchronous psychosocial online counselling, it is crucial to monitor the timeline of client responses to maintain communication and support (see Fig. 2). Understanding when to wait for a reply or send a follow-up message can greatly enhance the counselling process. Recalling precise details and response patterns

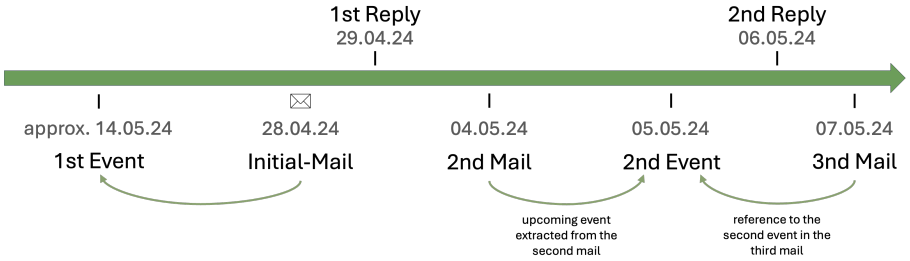

Fig. 2. Timeline feature in asynchronous psychosocial online counselling. The timeline tracks key dates and events in the communication between a client and counsellor. Counsellor's replies are displayed above the timeline, while events from the client are shown below the timeline. The green arrows indicate the source of the information, linking the events to their corresponding context within the communication history.

is challenging, particularly when communication is delayed. It can be unclear whether a client's lack of response is typical or indicative of an underlying issue. Additionally, if a client does not respond after mentioning an important event, it may be necessary to consider sending a follow-up message. Misunderstandings or confusing events can disrupt the relationship of trust and the flow of information between client and counsellor.

The Timeline feature employs the use of LLMs and straightforward algorithms to analyse the entirety of the email history, subsequently generating a comprehensive timeline. The timeline encompasses the dates and times of all

communications and responses exchanged between the client and the counsellor, in addition to any noteworthy occurrences that were mentioned during the sessions. The process begins with the application of elementary algorithms to extract temporal data from the email history, thereby identifying the dates and times at which each message was transmitted and received by the counsellor. Subsequently, the email conversations are presented to the LLM with a meticulously designed prompt, which instructs the LLM to identify and extract pertinent events in JSON format. The resulting output should comprise the name of each extracted event, the specific text in which the event is mentioned, and the source email from which the information was extracted. The output is validated to ensure that the extracted and cited textual elements are present in the original data and align with the specified email in the output. The validated data is then employed to construct a comprehensive timeline. The timeline is updated with every new email to ensure that it is up to date.

Developing the Timeline feature involves several significant challenges, particularly in training AI systems to accurately extract and organize information from email communications. One primary issue is the potential length of email threads, which can exceed the context window of the LLM. Similar to the approach used in the Case Summary feature, the solution involves iteratively removing the oldest emails until the current timeline and the most recent emails fit within the context window. This ensures that all essential information is included without exceeding the model's processing capabilities.

Another challenge is defining what constitutes a relevant event. While LLMs can already extract events and their corresponding text passages, determining the relevance of these events requires special instructions or fine-tuning on labeled data from email histories.

3.4 Social Network Graph in Psychosocial Online Counselling

In psychosocial online counselling, clients often reveal new information about their social networks gradually over time. As these details emerge, it becomes challenging for counsellors to maintain a comprehensive overview of all relevant stakeholders and their interrelationships. The individuals mentioned in these communications have relationships with the client and also with each other, creating a complex social construct that can be difficult to interpret solely through email exchanges. To address this challenge, a Social Network Graph (SNG) can be utilized (see Fig. 3).

The SNG displays individual actors within a client's social network and their relationships, enabling an automated interpretation of the client's social environment. This visual representation aids counsellors in comprehending the client's social context.

The primary issue in managing a client's social network during online counselling sessions is the gradual and fragmented disclosure of information. As clients provide new details over time, the list of stakeholders grows, making it difficult to maintain an accurate and comprehensive view. Counsellors need a method to

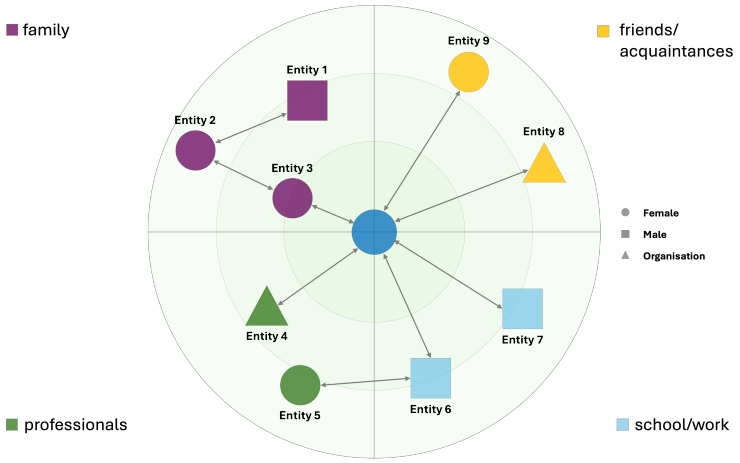

Fig. 3. This illustration shows an example SNG visualizing the different entities within a client's social environment and their interrelationships, including family members, friends, and professional contacts. Each entity is represented by different shapes and colors, as indicated in the legends. The four sectors represent different areas of the client's life, and the arrows indicate social connections between entities. The client is always positioned at the center of the graph.

visualize and organize this information effectively to ensure they understand the client's social dynamics and can provide informed support.

While an LLM could be used to extract individuals, their attributes and relationships from the complete email history, a step-by-step approach using multiple models offers more control and makes errors easier to identify and correct (Fig. 4 illustrates our approach). The process begins with the extraction of all named persons using named-entity recognition. Different mentions of the same person are merged to ensure that name variations are consolidated. Building on the results of the named person extraction, three parallel work packages are initiated. Each of these packages uses the email history and the already extracted person entities as their data basis. First, attributes such as age, gender, and alive/deceased status are extracted in JSON format. Any errors are flagged and corrected through feedback loops. The second parallel work package focuses on relationship extraction. Here, the LLM identifies the relationships between the entities and their coreferences. The output is generated again in JSON format, detailing these relationships. Once a relationship between two person entities is identified, it is depicted in the final visualization with a bidirectional arrow, indicating mutual connections. The third parallel work package involves the categorization of person entities into one of four sectors: family, professionals, friends/acquaintances, and school/work. Initially, this categorization can be performed by an LLM due to its ability to determine sectors from context with a well-crafted prompt and JSON output. As more data becomes available, an encoder model like BERT can be employed to classify entities more

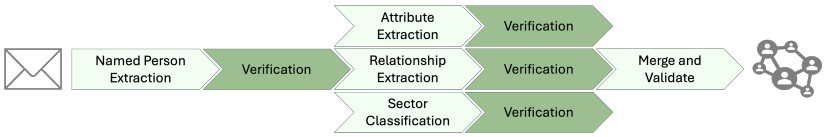

Fig. 4. This illustration shows the pipeline for creating a SNG. The process starts with Named Entity Recognition and verification, followed by three parallel work packages: Attribute Extraction, Relationship Extraction, and Sector Classification. Finally, all data is merged and validated, resulting in a comprehensive JSON file used to create the SNG.

resource-efficiently. Immediately following every step, a reasoner is integrated to verify the output. After verification and any necessary iterative loops in the individual work packages are completed, all the data is merged and saved as a JSON file. This comprehensive JSON output can then be plotted to create the SNG. This graph provides a clear and up-to-date visualization of the client's social environment. Once an SNG is generated, it is provided in the pipeline so that subsequent analyses do not need to start from scratch, thereby avoiding the repetition of previously made errors. Similar to the Case Summary and Timeline Management, if the email history exceeds the context window of the used LLM, older messages are discarded.

3.5 Case Classification with AI Support

The categorisation of cases within counselling organisations is based on the type of advisory request. This process is necessary for two main reasons: to ensure that cases are assigned to the required counsellors with the relevant expertise and to enable the compilation of meaningful statistics. Categorisation takes place in two phases: an initial and a final categorisation. This is necessary because the category can change in the course of counselling. Categorisation enables the allocation of clients to the most suitable counsellors. At the same time, statistical data is necessary to identify patterns and trends and to facilitate organisational planning.

The primary challenge lies in the accurate and efficient classification of cases into appropriate categories. This process is time-consuming for the counsellor, particularly if there has been a prolonged period of inactivity or if the case needs to be reviewed after a significant delay.

To address this challenge, an initial categorisation can be performed using an LLM to classify cases based on context. A well-crafted prompt containing the categories provides clear instructions for outputting the appropriate category. Post-processing can then extract the single category from the response. Once a sufficient quantity of data has been accumulated, a classifier can be trained. Fine-tuned encoder-only models like BERT still achieve stated-of-the-art performance for classification tasks, but are much more resource-efficient and faster than LLMs.

The primary challenge for supervised training is the acquisition of a substantial and heterogeneous corpus of labeled data. Once a sufficient quantity of data is available, the model can be trained and evaluated in an efficient and straightforward manner, and the LLM can be replaced by an encoder model.

4 Ethical Considerations

In the context of integrating AI into decision-making processes, several key ethical considerations must be highlighted. The primary focus should be on data bias, which can arise from unbalanced datasets that overrepresent certain demographic groups or conditions [3, 15]. This can lead to models making biased decisions, perpetuating stereotypes or systemic inequalities [29]. Such biases underscore the critical need for fair and representative training data to avoid unfair outcomes. Another significant concern is the potential loss of meaningful human control [7]. AI systems, despite their advanced capabilities, lack the moral consciousness and empathy inherent to human decision-making. While AI can assist in identifying patterns and providing recommendations, it cannot replace the nuanced judgment required in ethical deliberations, particularly in complex, context-sensitive situations. It is crucial that human professionals retain ultimate decision-making authority, ensuring that AI's role is limited to augmenting rather than replacing human judgment. This can be achieved through comprehensive training of counsellors, ensuring they understand when and how to integrate AI insights with their professional judgment. The ethical reflection on automation bias is also essential. This bias can occur when human professionals uncritically accept the outputs of AI systems, assuming their accuracy without sufficient scrutiny [18]. This issue is compounded by the complexity of AI models, which can be perceived as "black boxes" due to their opaque decision-making processes [7]. Ensuring transparency in how these systems operate and making their reasoning accessible to human users is vital for maintaining trust and accountability. This can be supported through the training of counsellors, helping them understand the underlying algorithms and the importance of questioning AI outputs.

Additionally, the ethical design and deployment of AI systems require continuous quality management and ethical training for the professionals using them [6]. This includes understanding the limitations and capabilities of AI technologies, fostering a culture of critical engagement, and ensuring that AI tools are used to support, rather than dictate, professional practices.

All features have been designed with these ethical considerations in mind. The Automatic Subject Generation may introduce bias towards certain keywords or phrases, but since it only affects the initial message, the impact is less critical, and errors are quickly recognizable. The Case Summarisation may present issues of data distortion, potentially leading to incomplete or hallucinated summaries. This could result in automation bias if counsellors rely too heavily on these summaries without sufficient scrutiny. To mitigate this, counsellors must receive ethical training to recognize and address such biases. Timeline Man-

agement is designed to minimize critical errors by using rule-based extraction of factual data, such as email timestamps. Although the importance of certain events may be misjudged, the structured nature of timelines ensures that events are logged accurately. This feature includes post-processing checks to validate the extraction of events. The SNG can be affected by biases in the inclusion and categorization of actors within a client's social network. Errors such as missing or duplicated actors may arise, but they can be identified through critical review. This feature offers the possibility of editing and enables counsellor to actively engage with the social circumstances of the client. Case Classification is another area where historical biases in the data can influence outcomes, potentially leading to stereotypical classifications. To address this, it is essential to maintain a balanced dataset, minimizing bias and enhancing classification fairness. Continuous monitoring and refinement of the classification models are necessary to ensure that they accurately reflect the diversity of cases and do not perpetuate existing biases.

In conclusion, the ethical integration of AI into psychosocial email counselling requires careful consideration of data privacy, bias mitigation, human control, ethical training, transparency, and the irreplaceable role of human empathy. Addressing these concerns ensures AI effectively supports and enhances the work of human counsellors, providing high-quality mental health services.

5 Conclusion and Future Work

The integration of AI, particularly LLMs, into psychosocial online counselling offers significant potential for enhancing case management, data visualization, and tracking client interactions. These AI-powered tools can map client networks and categorize cases, streamline and condense information, potentially reducing administrative burdens and allowing counsellors to focus more on personalised client support. However, it is essential to address the ethical considerations discussed earlier carefully. The development and deployment of AI in this domain must involve a proactive and dynamic approach to these ethical challenges, ensuring responsible integration. By doing so, AI can serve as a supportive tool that enhances the work of human counsellors rather than replacing their essential functions.

The CAIA project is currently in an active development phase, during which the first prototype was evaluated in collaboration with a partner organization for parent-child counselling. Usability tests and workshops were conducted to gather first user feedback. These initial evaluations confirmed the usefulness of the already implemented features and led to the conception of new ideas. For example, it is common for counsellors to consult with colleagues about a case, focusing on discussion and idea generation. A virtual colleague function could facilitate this process by acting as a knowledgeable partner familiar with the entire case and all features. This virtual colleague would enable counsellors to delve into the case in depth and consider various perspectives, leveraging the AI's comprehensive understanding of the situation. Another desired function, hypothesis

generation, is supposed to reflect on the client's messages and underlying issues, generating hypotheses about the reasons behind specific problems or concerns and their potential connections. A potential feature could provide suggestions for the counsellor's reply by analyzing previous communications, offering recommendations that align with the counsellor's expertise and the specifics of the case. In addition, the ability to highlight specific information within the email correspondence and jump directly from the synthesized information in the feature to the relevant passage in the original email would be nice. The AI must be implemented in such a way that it consistently indicates the source of the information displayed and ensures the traceability of each data point to the original content. This feature would allow advisors to accurately verify and contextualise the information.

To develop these features, it is crucial to collect anonymised data to train and evaluate AI models, ensuring they are fair and representative. Diverse data helps create balanced AI systems suited to a wide range of cases. Additionally, selecting appropriate LLMs for mental health counselling and fine-tuning them for specific needs is essential. A comprehensive operational concept must also be developed for securely using LLMs, ensuring strict data protection standards.

Future tests with the partner organization are planned, during which the presented features will be evaluated by real counsellors over an extended period. This prolonged testing phase aims to continuously develop CAIA according to the practical needs of psychosocial counsellors, ultimately enhancing the quality and efficiency of the support provided.

By continually refining AI systems, tools can enhance the quality of care in psychosocial online counselling. CAIA is designed to support counsellors ethically, without undermining their expertise or decision-making authority. The AI assists rather than replaces, enabling counsellors to engage deeply with each case and gain comprehensive insights. This approach allows for more personalised and effective client support. With ethical implementation, AI can significantly aid human counsellors, enhancing their capabilities and providing high-quality mental health services.

References

1. Ayers, J.W., Poliak, A., Dredze, M., Leas, E.C., Zhu, Z., Kelley, J.B., Faix, D.J., Goodman, A.M., Longhurst, C.A., Hogarth, M., Smith, D.M.: Comparing Physician and Artificial Intelligence Chatbot Responses to Patient Questions Posted to a Public Social Media Forum. JAMA Internal Medicine (2023)
2. Baldofski, S., Kohls, E., Efe, Z., Eckert, M., Saee, S., Thomas, J., Wundrack, R., Rummel-Kluge, C.: The Impact of a Messenger-Based Psychosocial Chat Counseling Service on Further Help-Seeking Among Children and Young Adults: Longitudinal Study. JMIR Mental Health (2023)
3. Belenguer, L.: AI bias: exploring discriminatory algorithmic decision-making models and the application of possible machine-centric solutions adapted from the pharmaceutical industry. AI and Ethics (2022)

4. Boucher, E.M., Harake, N.R., Ward, H.E., Stoeckl, S.E., Vargas, J., Minkel, J., Parks, A.C., Zilca, R.: Artificially intelligent chatbots in digital mental health interventions: a review. Expert Review of Medical Devices (2021)
5. Brocki, L., Dyer, G.C., Gładka, A., Chung, N.C.: Deep Learning Mental Health Dialogue System. 2023 IEEE International Conference on Big Data and Smart Computing (BigComp) (2023)
6. Campos Zabala, F.J.: Responsible AI Understanding the Ethical and Regulatory Implications of AI. Grow Your Business with AI: A First Principles Approach for Scaling Artificial Intelligence in the Enterprise (2023)
7. Cavalcante Siebert, L., Lupetti, M.L., Aizenberg, E., Beckers, N., Zgonnikov, A., Veluwenkamp, H., Abbink, D., Giaccardi, E., Houben, G.J., Jonker, C.M., van den Hoven, J., Forster, D., Lagendijk, R.L.: Meaningful human control: actionable properties for AI system development. AI and Ethics (2023)
8. Chen, S.: Nonverbal Communication in Intercultural Communication: A Case Study of Telling Lies. Frontiers in Educational Research (2023)
9. Chen, S., Wu, M., Zhu, K.Q., Lan, K., Zhang, Z., Cui, L.: LLM-empowered Chatbots for Psychiatrist and Patient Simulation: Application and Evaluation (2023)
10. Chen, X., Wang, T., Zhu, Q., Guo, T., Gao, S., Lu, Z., Gao, X., Zhang, X.: Rethinking Scientific Summarization Evaluation: Grounding Explainable Metrics on Facet-aware Benchmark (2024)
11. Chiu, Y.Y., Sharma, A., Lin, I.W., Althoff, T.: A Computational Framework for Behavioral Assessment of LLM Therapists (2024)
12. Chung, N.C., Dyer, G., Brocki, L.: Challenges of Large Language Models for Mental Health Counseling (2023)
13. D'Alfonso, S.: AI in mental health. Current Opinion in Psychology (2020)
14. D'Alfonso, S., Santesteban-Echarri, O., Rice, S., Wadley, G., Lederman, R., Miles, C., Gleeson, J., Alvarez-Jimenez, M.: Artificial Intelligence-Assisted Online Social Therapy for Youth Mental Health. Frontiers in Psychology (2017)
15. Ferrara, E.: Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies. Sci (2024)
16. Fu, G., Zhao, Q., Li, J., Luo, D., Song, C., Zhai, W., Liu, S., Wang, F., Wang, Y., Cheng, L., Zhang, J., Yang, B.X.: Enhancing Psychological Counseling with Large Language Model: A Multifaceted Decision-Support System for Non-Professionals (2023)
17. Geraldina, A.M., Suen, M.W., Suanrueang, P.: Online mental health services during COVID-19 pandemic in Indonesia: Challenges from psychologist perspective. PLOS ONE (2023)
18. Goddard, K., Roudsari, A., Wyatt, J.C.: Automation bias: a systematic review of frequency, effect mediators, and mitigators. Journal of the American Medical Informatics Association : JAMIA (2012)
19. Hsu, S.L., Shah, R.S., Senthil, P., Ashktorab, Z., Dugan, C., Geyer, W., Yang, D.: Helping the Helper: Supporting Peer Counselors via AI-Empowered Practice and Feedback (2023)
20. Java, S., Mohammed, H., Bhardwaj, A.B.: Psychological Analysis of Online Counselling Platforms Offering Mental Health Support. 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT) (2021)
21. Kiriakaki, S., Tzovanou, M., Sotiriou, A., Lampou, M., Varsamidou, E.: Online Counseling: Advantages, Disadvantages and Ethical Issues. Homo Virtualis (2022)

22. Korostiy, V., Missmahl, I., Polishchuk, O., Platinuk, O.: Psychosocial online counselling in Ukraine on IPSO-care platform in pandemic period. European Psychiatry (2021)
23. Koutsouleris, N., Hauser, T.U., Skvortsova, V., De Choudhury, M.: From promise to practice: towards the realisation of AI-informed mental health care. The Lancet Digital Health (2022)
24. Lai, T., Shi, Y., Du, Z., Wu, J., Fu, K., Dou, Y., Wang, Z.: Psy-LLM: Scaling up Global Mental Health Psychological Services with AI-based Large Language Models (2023)
25. Liu, J.M., Li, D., Cao, H., Ren, T., Liao, Z., Wu, J.: ChatCounselor: A Large Language Models for Mental Health Support (2023)
26. Liu, L., Liu, Y.: An Research on Online Counseling Platform Based on the Artificial Intelligence Technology. 2022 2nd International Conference on Bioinformatics and Intelligent Computing (2022)
27. Maurya, R.K.: Using AI Based Chatbot ChatGPT for Practicing Counseling Skills Through Role-Play. Journal of Creativity in Mental Health (2023)
28. Maurya, R.K., DeDiego, A.C.: Artificial intelligence integration in counsellor education and supervision: A roadmap for future directions and research inquiries. Counselling and Psychotherapy Research (2023)
29. Morales, S., Clarisó, R., Cabot, J.: Automating Bias Testing of LLMs. 2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE) (2023)
30. Mulungu, C., Mindu, T., Mulungu, K.: Effectiveness of online counselling during COVID-19 in Zambia. A client and therapists perspective. BMC psychology (2024)
31. Rudolph, E., Engert, N., Albrecht, J.: An AI-Based Virtual Client for Educational Role-Playing in the Training of Online Counselors:. Proceedings of the 16th International Conference on Computer Supported Education (2024)
32. Shah, H.: A.L.I.C.E.: an ACE in Digitaland. tripleC: Communication, Capitalism & Critique. Open Access Journal for a Global Sustainable Information Society (2006)
33. Sharma, A., Lin, I.W., Miner, A.S., Atkins, D.C., Althoff, T.: Towards Facilitating Empathic Conversations in Online Mental Health Support: A Reinforcement Learning Approach (2021)
34. Sharma, A., Lin, I.W., Miner, A.S., Atkins, D.C., Althoff, T.: Human-AI Collaboration Enables More Empathic Conversations in Text-based Peer-to-Peer Mental Health Support (2022)
35. da Silva, J.A.M., Siegmund, G., Bredemeier, J.: Crisis interventions in online psychological counseling. Trends in Psychiatry and Psychotherapy (2015)
36. Situmorang, D.D.B.: Online/Cyber Counseling Services in the COVID-19 Outbreak: Are They Really New? The Journal of Pastoral Care & Counseling (2020)
37. Stoll, J., Müller, J.A., Trachsel, M.: Ethical Issues in Online Psychotherapy: A Narrative Review. Frontiers in Psychiatry (2020)
38. Tuna, B., Avci, O.H.: Qualitative analysis of university counselors' online counseling experiences during the COVID-19 pandemic. Current Psychology (2023)
39. Weizenbaum, J.: ELIZA—a computer program for the study of natural language communication between man and machine. Communications of the ACM (1966)
40. Xu, B., Zhuang, Z.: Survey on psychotherapy chatbots. Concurrency and Computation: Practice and Experience (2022)